

Feasibility-Aware Quantum-Classical Benchmarking of a HUBO Formulation for Occlusion-Aware UAV Trajectory Optimization

Abstract—Trajectory optimization for unmanned aerial vehicle (UAV) target tracking in cluttered environments must simultaneously satisfy obstacle-avoidance and *occlusion-free visibility* (line-of-sight) constraints. This paper presents a discrete Higher-Order Unconstrained Binary Optimization (HUBO) formulation for occlusion-aware UAV trajectory optimization, designed to enable benchmarking against classical heuristics, against simulated-annealing (SA) samplers (after standard quadratization to QUBO), and against the Quantum Approximate Optimization Algorithm (QAOA) on a reduced instance. We further introduce a *feasibility-aware* evaluation methodology that decouples hard-constraint satisfaction from soft trajectory quality, applies independent constraint verification, matches sampling budgets, and characterizes penalty-weight exploitation. On an 8×8 instance (960 binary variables, 87,684 monomials, 3,172 cubic terms), we compare a classical A* baseline; a custom trajectory-level SA that evaluates the HUBO *natively*; the `neal` classical SA sampler [13] applied via `dimod.HigherOrderComposite` [14] which quadratizes the HUBO into a 1,542-variable QUBO (+582 ancillas, +60 %); and Qiskit QAOA [15] on a reduced 8-qubit QUBO. A* and custom SA reach comparable soft cost (317 vs. 303); `neal` attains only 22 % true feasibility, increasing from 0 % at $\lambda = 50$ to 60 % at $\lambda = 2000$. QAOA reproduces the optimum on the reduced QUBO as a correctness check.

Index Terms—HUBO, QUBO, quadratization, simulated annealing, QAOA, occlusion-aware tracking, UAV trajectory, A*, feasibility-aware benchmarking.

I. INTRODUCTION

Problem. A quadrotor must traverse a discretized 2D workspace from a fixed start cell to a target cell while (i) avoiding obstacle cells, (ii) respecting 4-connected motion over a finite horizon T , and (iii) preferring cells from which the target is not occluded by obstacles along the line of sight. A cell short in Euclidean distance can incur a large *occlusion* penalty if intervening obstacles block the line of sight, making this strictly more than shortest-path planning.

Terminology. *Obstacle-avoidance*: UAV cannot occupy obstacle cells. *Visibility-aware* (= *occlusion-aware*): trajectories are penalized in proportion to the number of obstacle cells lying on the line-of-sight from the UAV to the target. *Feasibility-aware* (our methodology): solver quality is judged only after separating hard from soft cost and only on independently-verified-feasible samples.

Related work. Visibility-aware UAV target tracking has been treated in *continuous* MPC form by Chen *et al.* [1], Hausman *et al.* [2], Penin *et al.* [3], Falanga *et al.* [4] (PAMPC), Wang *et al.* [5], and Masnavi *et al.* [6] (multi-convex MPC, the specific problem this paper reformulates discretely). In the

quantum-optimization setting, QUAUV [7] addresses obstacle-only UAV planning with QAOA and Q4DR [8] addresses drone routing; neither treats visibility-constrained trajectory planning.

Scope and contributions. The contribution is not a claim of quantum speedup, but the introduction of a quantum-compatible HUBO benchmark and a feasibility-aware methodology. Classical baselines are deliberate: current hardware limits make strong classical references essential, while the higher-order formulation and reduced-instance QAOA establish a path toward future QUBO/QAOA and D-Wave hardware studies. Specifically: (1) a discrete HUBO reformulation of the Masnavi class, bridging continuous MPC and higher-order binary optimization; (2) a feasibility-aware methodology with a classical A* baseline; (3) an analysis of why the cubic sustained-proximity term requires auxiliary variables for quadratization, with measured auxiliary count for our instance.

II. HUBO REFORMULATION

Let V be the set of grid cells, T the horizon. Binary $x[t, v] \in \{0, 1\}$ is 1 iff the UAV occupies cell v at step t . The Hamiltonian has seven terms:

$$H = \underbrace{H_{\text{uniq}} + H_{\text{start}} + H_{\text{move}} + H_{\text{obs}}}_{\text{hard constraints}} + \underbrace{H_{\text{occ}} + H_{\text{prox}} + H_{\text{term}}}_{\text{soft cost}}. \quad (1)$$

Hard terms enforce one-hot occupancy (H_{uniq}), fixed start (H_{start}), 4-connected motion (H_{move}), and obstacle exclusion $H_{\text{obs}} = \lambda_{\text{obs}} \sum_t \sum_{v \in O} x[t, v]$, with $O \subset V$ the obstacle set. H_{term} rewards reaching the goal cell.

Occlusion penalty. For each cell u we trace a Bresenham line of sight to the target u_{tgt} ; let $\text{LOS}(u, u_{\text{tgt}}) \subset V$ be the set of cells strictly between them.

$$H_{\text{occ}} = \lambda_{\text{occ}} \sum_t \sum_{u \in V} x[t, u] |\text{LOS}(u, u_{\text{tgt}}) \cap O|, \quad (2)$$

i.e. a per-cell penalty equal to the number of obstacles blocking the target (Fig. 1a). Linear in x for a static target; intrinsically higher-order for a *moving* target whose position is also a decision variable.

Genuinely cubic proximity. Let $B \subset V$ be the *buffer cells* – non-obstacle cells adjacent to an obstacle. The sustained-

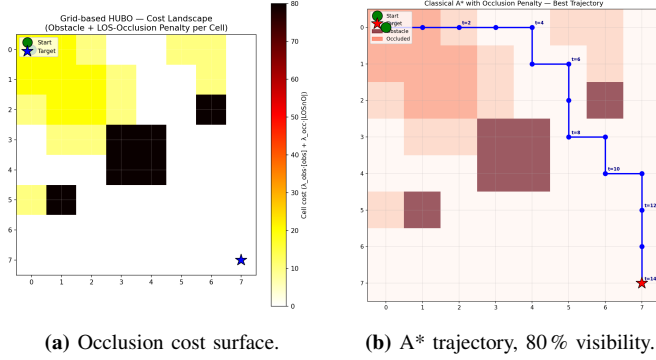


Fig. 1: (a) Per-cell obstacle + line-of-sight occlusion cost to the target; warm cells are occluded, dark cells obstacles. (b) Best A* trajectory: feasible, target-reaching.

proximity term penalizes lingering in a buffer cell for three consecutive steps:

$$H_{\text{prox}} = \lambda_{\text{prox}} \sum_{v_b \in B} \sum_{t=1}^{T-1} x[t-1, v_b] x[t, v_b] x[t+1, v_b]. \quad (3)$$

Transient passage incurs no penalty; only lingering does. Each summand is a degree-three product.

Quadratization cost. Eq. (3) cannot be evaluated by QUBO-only samplers without first applying Rosenberg reduction [10], [11], which introduces an auxiliary variable $y_{ij}=x_i x_j$ per cubic product. Running `dimod.make_quadratic` [14] on our full HUBO yields 1,542 variables and 88,470 quadratic interactions (**+582 auxiliaries, +60 %**). The resulting dense couplings degrade the sparsity that annealing samplers exploit – a cost silently incurred whenever a higher-order problem is solved via a quadratic-only sampler.

III. FEASIBILITY-AWARE METHODOLOGY

A methodology is *feasibility-aware* when it decouples hard from soft cost at every evaluation step, because penalty-based solvers can lower aggregate (“raw”) energy by violating soft constraints. Safeguards: (i) energy decomposition into hard penalty ($H_{\text{uniq}} + H_{\text{start}} + H_{\text{move}} + H_{\text{obs}}$) and soft cost; (ii) independent constraint verification on decoded trajectories (not energy thresholds); (iii) matched sampling budgets (50 samples each); (iv) penalty-weight sweep over orders of magnitude to expose constraint exploitation.

Solvers. (a) **A*** [9]: cell cost $1 + \lambda_{\text{occ}} |\text{LOS} \cap O| + \lambda_{\text{prox}} \mathbf{1}[v_b \in B]$, Manhattan heuristic – an *approximate* HUBO-soft analogue (shares occlusion+proximity, encodes goal via search termination). (b) **Custom SA**: trajectory-level moves respecting connectivity; evaluates the HUBO *natively* (products of binary indicators including the cubic H_{prox}). (c) **neal** [13]: a *classical* SA sampler from D-Wave Ocean accepting only quadratic problems; applied to our HUBO via `dimod.HigherOrderComposite`, which quadratizes first. (d) **Qiskit QAOA** [12], [15]: state-vector simulator; its `QuadraticProgram` accepts only linear+quadratic terms.

TABLE I: Tier 1: 8×8 HUBO (960 binary variables). *neal* solves the quadratized QUBO with 582 ancillas (1,542 total); custom SA solves the HUBO natively.

Metric	A*	Custom SA (native HUBO)	neal (QUBO)
Best soft cost	317	303	496
Feasibility	100 %	100 %	22 %
Reaches target	yes	14 % runs	8 % runs
Visibility (% steps)	80 %	— [†]	— [†]
Runtime per sample	10 ms	47.8 s	0.26 s

[†]Per-step visibility not logged in the stochastic run.

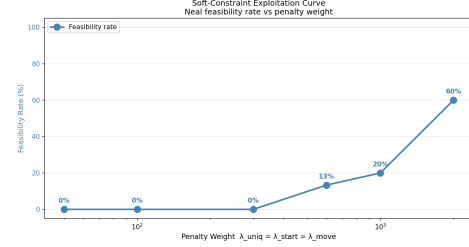


Fig. 2: Constraint-exploitation curve: *neal* feasibility vs. penalty weight λ . General-purpose SA samplers need penalty weights an order of magnitude above the objective scale to respect one-hot HUBO structure.

Two tiers. State-vector QAOA scales as 2^n , impractical at $n=960$. **Tier 1** (full HUBO): 8×8 , $T=15$, 960 vars, 3,172 cubic; A*, custom SA, *neal* (on its 1,542-var quadratized QUBO). **Tier 2** (QUBO correctness): 2×2 , $T=2$, no obstacles, **8 binary variables / 8 qubits**, already linear+quadratic (no buffer cells, so the cubic term trivializes). 50 matched samples per stochastic solver, SA 500 sweeps, $\lambda=1000$.

IV. RESULTS

Tier 1. Table I: A* and custom SA achieve comparable soft cost (317 vs. 303); A* completes in 10 ms, SA in 47.8 s, both feasible. *neal* (on the quadratized QUBO) is fast (0.26 s) but only 22 % feasible; its best feasible soft cost (496) is 56 % worse than A*’s (Fig. 1b).

Exploitation curve. Sweeping λ (Fig. 2): *neal* feasibility climbs from 0 % ($\lambda \leq 300$) to 13 %, 20 %, 60 % at $\lambda=600, 1000, 2000$. A single λ would mis-state *neal*’s capability; only the sweep exposes the exploitation regime.

Tier 2. On the 8-var QUBO (no cubic), custom SA, *neal*, and Qiskit QAOA (SPSA, reps=1, maxiter=25, 8 qubits) all reach $E^*=-118.17$. Runtime: QAOA 55 s, *neal* 0.02 s, SA 0.14 s. This validates encoding-to-QAOA-readiness; QAOA scales exponentially in qubits and is impractical at Tier-1 scale.

V. DISCUSSION AND CONCLUSION

The HUBO captures sensible planning structure: A* finds a feasible trajectory in ms within 5 % of the best stochastic soft cost. The HUBO’s value here is (i) compatibility with quantum-style samplers via documented quadratization (+582 ancillas, +60 %), and (ii) extensibility to a *moving* target, where Eq. (2) becomes intrinsically cubic, beyond A*’s reach. Feasibility-aware benchmarking proves necessary: 78 % of *neal*’s hardened-weight samples violate constraints. Future work: moving target; real D-Wave Advantage hardware (embedding the 1,542-var QUBO with further minor-embedding overhead); per-step visibility for SA/*neal*; comparison vs. multi-convex MPC [6]. *Thanks to all co-authors.*

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